

StyleMate – An AI-Powered Virtual Assistant for Personalized Clothing Selection Using 3D Body Modeling

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Declaration

We, the undersigned members of research group 25-26J-302, declare that this Final Group Report is the result of our own collective effort. The work documented in this report has been carried out in fulfilment of the requirements for the degree of B.Sc. (Hons) in Information Technology Specializing in Information Technology at the Sri Lanka Institute of Information Technology.

All sources of information used are acknowledged through standard in-text citations and a complete list of references at the end of this document. We declare that this report has not been previously submitted, in whole or in part, for any other award at this or any other institution.

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The above candidates have carried out research for the Final Group Report under my supervision and have submitted their work in conformity with the project guidelines.

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Acknowledgement

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Abstract

The StyleMate system is an AI-powered virtual fashion assistant designed to provide users with a personalized, context-aware, and inclusive clothing selection experience. The rapid growth of e-commerce has created a demand for intelligent virtual styling tools that go beyond simple product recommendations, addressing the real-world challenges of outfit planning, accessory selection, wardrobe management, and event-appropriate dressing.

This group project integrates four tightly coupled modules into a unified full-stack platform. The first module, the Accessory Recommendation and Customization System, employs computer vision to extract wearer attributes such as body shape, face shape, dress color, neckline style, and skin tone from user-uploaded images, and uses a rule-based engine to suggest complementary accessories alongside a user-driven design customization feature. The second module, the Smart Digital Wardrobe, allows users to digitize their existing clothing items through image recognition, generating context-aware outfit recommendations based on weather conditions, user occupation, and travel needs. The third module focuses on 3D Virtual Mannequin Generation, creating personalized digital avatars from standard 2D photographs using pose estimation, body shape classification, and skin tone analysis. The fourth module, the Event Theme Color Analysis and Outfit Recommendation System, analyzes uploaded event decoration images using K-Means clustering to extract dominant color palettes, which are then mapped to suitable outfit suggestions using color theory principles and occasion-based filtering.

All modules share a common React.js frontend and Flask/FastAPI backend architecture, enabling seamless data integration and a consistent user experience. Together, these components create a comprehensive and inclusive virtual styling ecosystem that improves personalization, reduces decision fatigue, and supports sustainable fashion practices.

Keywords: Virtual Fashion Assistant, AI Outfit Recommendation, Computer Vision, 3D Mannequin, Color Analysis, Accessory Recommendation, Digital Wardrobe, Event Theme, Personalized Styling, Inclusive Fashion Technology

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List of Abbreviations

Abbreviation	Full Form
AI	Artificial Intelligence
ML	Machine Learning
CV	Computer Vision
CNN	Convolutional Neural Network
GAN	Generative Adversarial Network
NLP	Natural Language Processing
API	Application Programming Interface
UI	User Interface
UX	User Experience
RGB	Red, Green, Blue
HEX	Hexadecimal Color Code
EDA	Exploratory Data Analysis
VTO	Virtual Try-On
3D	Three-Dimensional
2D	Two-Dimensional
HSV	Hue, Saturation, Value
SVM	Support Vector Machine
MOS	Mean Opinion Score
SUS	System Usability Scale
WCAG	Web Content Accessibility Guidelines
IDE	Integrated Development Environment
SRS	Software Requirements Specification

Chapter 1: Introduction

1.1 Background and Context

The fashion and e-commerce industries have witnessed unprecedented growth in the digital era, fundamentally transforming the way individuals discover, evaluate, and purchase clothing. With the global fashion e-commerce market expanding rapidly, consumers increasingly rely on online platforms to make clothing and styling decisions. However, existing virtual fashion systems have failed to replicate the rich, personalized, and contextual experience of in-store styling.

Traditional fashion recommendation systems operate primarily on product-centric data, suggesting new clothing items based on user purchase history, popularity trends, or demographic filters. These systems do not take into account the full spectrum of factors that govern real-world fashion decisions, including body shape and skin tone, existing wardrobe inventory, event theme and occasion type, or desired accessory coordination. As a result, users frequently encounter irrelevant suggestions, high product return rates, and a general lack of personalization.

The StyleMate project addresses these shortcomings by introducing a comprehensive, AI-powered virtual fashion assistant that integrates four distinct but interoperable modules. Each module targets a specific dimension of the fashion experience: accessory styling, wardrobe management, 3D avatar generation, and event theme-based outfit recommendation. Together, they form a unified system capable of delivering intelligent, inclusive, and context-aware fashion guidance.

This research is grounded in the convergence of computer vision, machine learning, natural language processing, and web technologies. By combining image-based analysis with contextual reasoning, StyleMate moves beyond transactional product recommendation towards holistic styling intelligence.

1.2 Literature Review

A substantial body of research has been dedicated to fashion recommendation and virtual try-on technology. Early virtual dressing room systems relied on basic 2D image overlays on static models, which lacked the ability to reflect individual body characteristics. More recent approaches have leveraged deep learning, particularly Convolutional Neural Networks (CNNs) and Generative Adversarial Networks (GANs), to achieve more realistic garment fitting and style matching.

Studies have demonstrated that visual features such as color, texture, and pattern play a critical role in determining fashion compatibility. Research on personalized outfit recommendation systems has shown that incorporating user-specific attributes, such as body measurements, skin tone, and style preferences, significantly improves recommendation relevance. However, most commercial and academic systems focus on clothing items in isolation and do not integrate environmental context, user wardrobe history, or event-specific styling requirements.

Work on color theory and harmony in digital systems has explored the use of complementary, analogous, and triadic color schemes for aesthetic matching. Clustering algorithms such as K-Means have been applied successfully to extract

dominant color palettes from images in domains including branding and interior design, though their application in fashion recommendation remains limited. Context-aware recommendation research in e-commerce has shown the value of incorporating external signals such as weather, occasion, and location, but few fashion systems have extended this to visual environmental analysis.

Research in affective computing and emotion-aware interfaces has identified the potential of incorporating user mood and emotional state into recommendation engines, improving engagement and satisfaction. Accessibility studies have highlighted significant gaps in virtual fashion tools for differently abled users, pointing to a need for flexible input modalities and inclusive interface design.

1.3 Research Gap

Despite the growing sophistication of AI-driven fashion systems, several critical gaps persist across the existing literature and commercial solutions.

First, existing virtual try-on systems depend on specialized hardware such as 3D scanners or multi-view camera rigs, making them inaccessible to general consumers. Systems that operate on standard 2D images typically require rigid, full-body, front-facing poses, excluding seated, partially visible, or differently abled users.

Second, most fashion recommendation systems are product-centric, encouraging new purchases rather than optimizing the user's existing wardrobe. Features such as mix-and-match outfit generation, travel wardrobe planning, and group or couple outfit coordination are rarely integrated into a single platform.

Third, while color theory is well established in design and fashion domains, its automated application in recommendation systems is superficial. Very few systems analyze external environmental images, such as event decoration photographs, to derive outfit color recommendations.

Fourth, accessory-specific recommendation systems are largely absent. Most platforms focus on clothing and neglect the role of accessories in completing a look, particularly recommendations tied to wearer-specific attributes such as neckline style, face shape, and skin tone.

Fifth, user-driven customization in fashion AI is underdeveloped. Most systems generate recommendations without allowing users to specify preferred designs, patterns, or styles for accessories or clothing items.

The StyleMate system directly addresses all five of these gaps through its four integrated modules, collectively delivering a novel, inclusive, and practically oriented virtual fashion assistant.

1.4 Problem Statement

The central problem addressed by this research is the absence of a comprehensive AI-powered virtual fashion assistant that can simultaneously address the personalization, accessibility, and contextual dimensions of clothing selection. Users face difficulties in selecting appropriate outfits for different events, coordinating accessories with their clothing, managing and maximizing the use of their existing wardrobe, and visualizing how clothing will look on their specific body type and skin tone. Existing systems address these challenges in isolation, and no

integrated platform currently provides a unified solution that combines event theme analysis, wardrobe optimization, 3D visualization, and accessory customization.

1.5 Research Objectives

1.5.1 Main Objective

To design and implement an integrated AI-powered virtual fashion assistant that provides personalized, context-aware, and inclusive clothing and styling recommendations through four specialized modules covering accessory recommendation, wardrobe management, 3D avatar generation, and event theme color analysis.

1.5.2 Specific Objectives

- To develop a computer vision-based accessory recommendation and customization system that extracts wearer attributes from uploaded images and applies fashion styling rules for personalized accessory suggestions.
- To implement a smart digital wardrobe management system that recognizes clothing items from uploaded images and generates weather-aware, occasion-sensitive, and mix-and-match outfit recommendations.
- To build a 3D virtual mannequin generation module that creates personalized digital avatars from standard 2D photographs using body shape classification, pose estimation, and skin tone analysis.
- To construct an event theme color analysis and outfit recommendation module that extracts dominant color palettes from event decoration images and maps them to suitable outfit suggestions using color harmony principles.
- To integrate all four modules into a unified full-stack web application that delivers a seamless, real-time, and user-friendly virtual styling experience.

1.6 Scope and Limitations

The scope of this project covers the design, development, and evaluation of four AI-powered modules integrated into the StyleMate web application. The system targets users seeking personalized fashion guidance for everyday outfit planning, event dressing, and wardrobe optimization. The platform is web-based and accessible via standard browsers on desktop and mobile devices.

Limitations include the dependency on image quality for computer vision tasks, the absence of physical try-on functionality, and the use of rule-based and clustering approaches rather than end-to-end deep learning for some modules due to computational resource constraints. The system is designed and evaluated primarily for a Sri Lankan user base, though its architecture supports internationalization.

1.7 Significance of the Research

This research contributes to the field of AI-driven fashion technology by presenting an integrated system that combines multiple specialized AI modules into a single, practically usable application. Unlike existing research that addresses individual

dimensions of fashion intelligence, StyleMate provides a holistic solution spanning accessory styling, wardrobe management, virtual body modeling, and environmental context analysis. The system promotes sustainable fashion practices by encouraging wardrobe reuse and reducing unnecessary purchases, and advances inclusivity in digital fashion by supporting diverse body types, skin tones, and user abilities.

Chapter 2: Methodology

2.1 System Overview

StyleMate is built as a modular, full-stack web application in which four independent but interoperable AI modules share a common frontend and backend infrastructure. The overall system architecture follows a client-server model, with a React.js frontend serving as the user interaction layer and a Python-based backend (Flask/FastAPI) handling image processing, AI inference, and recommendation generation.

Each module exposes a set of RESTful API endpoints consumed by the shared frontend. Data flows between modules where relevant, such as skin tone and body shape information from the 3D mannequin module being passed to the accessory recommendation engine. A shared MongoDB database stores user profiles, wardrobe inventories, and recommendation histories. The system is containerized using Docker and designed to be deployable on cloud infrastructure.

2.2 System Architecture

The high-level system architecture of StyleMate consists of the following layers:

- **User Interface Layer (React.js):** Provides the interactive frontend for all four modules. Users can upload images, select occasions, manage their wardrobe, and view recommendations in real time.
- **API Gateway Layer:** Routes requests from the frontend to the appropriate backend module. Handles authentication, session management, and request validation.
- **Module Processing Layer:** Contains the four independent AI processing pipelines, each with its own image analysis, feature extraction, and recommendation logic components.
- **Data Storage Layer (MongoDB):** Stores user profiles, clothing inventories, color palettes, accessory catalogs, and recommendation logs.
- **Deployment Layer (Docker/Cloud):** Packages each module as a containerized microservice for independent scaling and deployment.

2.3 Project Timeline

The project followed an Agile/SCRUM development methodology, with work divided into iterative sprints from July 2025 to June 2026. Key phases included Planning and Research (July–August 2025), Data Collection (August–September 2025), Data Preprocessing and Model Development (September–November 2025), Testing and Validation (November–December 2025), Application Integration and UI Design (December 2025–February 2026), Final Testing and Refinement (February–April 2026), and Launch and Monitoring (April–June 2026).

2.4 Software Development Methodology

The Agile methodology with SCRUM framework was adopted for the development of StyleMate. This approach enabled iterative and collaborative development, allowing each team member to work independently on their module while

maintaining regular integration checkpoints. Daily stand-ups, sprint reviews, and retrospectives were conducted to ensure continuous progress tracking and quality improvement. Version control was managed through Git/GitHub, and project management was tracked using Trello.

2.5 Tools, Libraries, and Technologies

The following technologies were used across the StyleMate system:

Category	Technology / Tool
Frontend	React.js, Material-UI (MUI), HTML5, CSS3, JavaScript
Backend	Python Flask, FastAPI, Node.js
Computer Vision	OpenCV, TensorFlow, PyTorch, MediaPipe
3D Modeling	Blender (Python API), Three.js (glTF/GLB rendering)
Database	MongoDB
Deployment	Docker, Kubernetes, Google Cloud Platform (GCP)
Version Control	Git (GitHub/GitLab)
UI Design	Figma
Project Management	Trello, Microsoft Teams
Testing	Pytest, Jest, Postman
Code Quality	SonarQube

Chapter 3: Component 1 – Accessory Recommendation & Customization System

Student: Vishnuyan T. (IT22208194)

3.1 Introduction

The Accessory Recommendation and Customization System is a specialized module within the StyleMate virtual fashion assistant that addresses a commonly overlooked dimension of personal styling: the intelligent recommendation and customization of accessories. Accessories play a fundamental role in completing and enhancing a fashion look, yet most virtual styling platforms focus exclusively on clothing and neglect this essential aspect.

The concept of a virtual dressing room has been driven by the increasing demand for personalized online shopping experiences. Traditional consumers relied on physical try-ons to evaluate clothing fit and accessory compatibility. The shift to online shopping has spurred the development of virtual platforms to replicate this experience. Early systems were limited to basic 2D representations, lacking the ability to account for wearer-specific attributes or personalized styling.

This module introduces a dual-approach solution: an automated recommendation engine that analyzes user attributes and applies fashion styling rules, and a user-driven customization feature that allows users to specify desired accessory designs. Together, these capabilities provide a more complete and expressive virtual styling experience.

3.2 Background and Literature Survey

Recent advancements in virtual fashion have focused on enhancing personalization through AI-driven systems. A framework for personalized outfit recommendations uses deep learning to analyze user images and extract attributes like clothing style and body shape, employing CNNs to suggest style-compatible outfits. While effective for clothing ensembles, this approach does not address accessory-specific recommendations or user-driven customization.

Personalized recommendation systems have also evolved, with models integrating visual features from product images with user preferences to suggest complementary fashion items. This approach excels in general fashion recommendations but does not incorporate wearer-specific attributes like face shape, neckline style, or skin tone, nor does it support user-defined customization.

Customization in virtual fashion has gained momentum through generative AI. A framework for generating custom clothing designs based on user text inputs uses cycle-consistent GANs to create novel patterns. However, this system focuses on apparel and lacks real-time attribute extraction from user images. A virtual styling system that analyzes user-uploaded images to recommend accessories based on clothing attributes exists but lacks a customization module, restricting user control over design preferences.

3.3 Research Gap

Current virtual fashion systems prioritize clothing over accessories, neglecting their role in completing a look. Most systems recommend clothing items based on visual and contextual features but overlook accessories tied to wearer attributes like neckline style or face shape. Existing personalized outfit generation systems leverage visual features and user preferences but fail to incorporate detailed attributes such as body shape or skin tone, reducing the precision and inclusivity of accessory recommendations.

Generative AI enables custom apparel designs from text prompts but lacks real-time attribute extraction from user images, which is critical for context-aware accessory recommendations. Existing systems also rarely allow users to request or modify accessory designs, limiting creative expression. There is a clear gap for an integrated system that combines real-time attribute extraction, precise accessory recommendations, and user-specified design generation or retrieval.

3.4 Research Problem

The rise of online shopping has increased the demand for virtual dressing room technologies that replicate the personalized experience of physical try-ons. Most virtual fashion platforms focus on clothing recommendations, overlooking the critical role of accessories. These systems often fail to integrate wearer-specific attributes such as body shape, face shape, dress color, neckline style, and skin tone, which are essential for precise and context-aware accessory recommendations. For example, recommending a rope chain necklace to complement a V-neck dress requires analyzing specific attributes that current systems do not adequately address. There is also a lack of user-driven customization options, limiting individuals' ability to request specific accessory designs.

3.5 Objectives

Main Objective: To develop an intelligent virtual dressing room component that delivers personalized, inclusive, and creative accessory styling solutions by integrating computer vision to extract wearer-specific attributes, a rule-based recommendation engine for precise accessory suggestions, and a customization module enabling user-requested design generation or retrieval.

Specific Objectives:

- Develop a computer vision module to accurately extract body shape, face shape, dress color, neckline style, and skin tone from user-uploaded full-body images.
- Design a rule-based recommendation engine to suggest complementary accessories based on extracted attributes, ensuring aesthetic compatibility across diverse user profiles.
- Implement a user-driven customization module allowing users to request specific accessory designs or patterns, generated or retrieved from a catalog.
- Ensure inclusivity and accessibility across diverse body shapes, skin tones, and cultural styling preferences.

- Integrate and evaluate system performance through user studies measuring accuracy, user satisfaction, and inclusivity.

3.6 Methodology

3.6.1 Requirement Gathering

Requirement gathering involved stakeholder interviews with fashion designers and e-commerce developers, user surveys quantifying preferences for accessory types and styling attributes, observation of users on existing platforms to identify usability issues, focus groups with diverse participants to explore customization needs, and literature and market analysis to inform technical requirements. Initial prototypes were tested to validate and refine requirements.

3.6.2 System Architecture

The Accessory Recommendation and Customization System architecture consists of three integrated layers. The Image Processing Layer receives user-uploaded full-body images and performs attribute extraction using TensorFlow-based CNNs and OpenCV, identifying body shape, face shape, dress color, neckline style, and skin tone. The Recommendation Engine Layer applies a comprehensive set of fashion styling rules to map extracted attributes to suitable accessory recommendations, such as suggesting rope chain necklaces for V-neck dresses. The Customization Module Layer uses PyTorch-based generative AI models to generate or retrieve user-specified accessory designs, such as heart-patterned necklaces, from an extensive MongoDB catalog.

3.6.3 Functional Requirements

- Image Upload: Allow users to upload full-body images via a web interface.
- Attribute Extraction: Extract body shape, face shape, dress color, neckline style, and skin tone using computer vision with 85% target accuracy.
- Accessory Recommendation: Suggest at least three inclusive accessory options based on extracted attributes.
- Customization: Enable users to request specific designs via text, generating or retrieving items accordingly.
- User Interface: Provide a responsive React JS and Material-UI interface for real-time visualization of recommendations.
- Data Management: Store user profiles and preferences in MongoDB, ensuring data privacy.

3.6.4 Tools and Technologies

Programming Language: Python. Frameworks: TensorFlow (CNN-based attribute extraction), OpenCV (image processing), PyTorch (generative AI for customization). Database: MongoDB. Frontend: React JS, Material-UI. Design: Figma. Version Control: Git (GitHub). Testing: Pytest, Postman. Deployment: Docker, Kubernetes. Project Management: Trello.

3.6.5 Testing Strategy

Unit testing with Pytest validates individual computer vision components targeting 85% accuracy across diverse images. Integration testing confirms cohesive

operation of the computer vision, recommendation, and customization modules through API endpoint testing with Postman. User acceptance testing involves diverse participants evaluating the interface for usability, inclusivity, and real-time visualization. Performance testing verifies responsiveness, and stress testing assesses stability under peak usage. Test cases ensure inclusivity across skin tones and cultural preferences.

3.7 Risk Management

Risk	Probability	Impact	Mitigation Strategy
Inaccurate Attribute Extraction	Medium (40%)	High	Robust preprocessing with OpenCV; train on diverse datasets; rigorous unit testing
Ineffective Recommendations	Medium (30%)	High	Expert-validated fashion styling rules; testing on diverse user profiles
Customization Module Failure	Low (20%)	Medium	PyTorch generative AI trained on diverse designs; extensive MongoDB catalog
Data Privacy Breach	Low (15%)	Critical	Image anonymization; encrypted MongoDB storage; security compliance testing
System Scalability Issues	Medium (35%)	High	Docker containerization; Kubernetes orchestration; stress testing

3.8 Commercialization

The target market includes Sri Lanka's fashion e-commerce market, valued at LKR 19.8 billion (2024), with rapid growth across Asia. The business model follows a freemium approach with a premium subscription (LKR 500/month) for personalized style insights and wardrobe features, alongside partnerships with online retailers and fashion brands for additional revenue via affiliate links. Development cost of approximately LKR 3.5 million can be recovered within 12-15 months by targeting 10,000 active users. Unique selling points include the first explainable fashion recommender fusing body and skin analysis, inclusivity for Asian body types and skin tones, and a complete styling ecosystem.

Chapter 4: Component 2 – Smart Digital Wardrobe

Student: Abirajh A. (IT22066848)

4.1 Introduction

The Smart Digital Wardrobe addresses the common problem of wardrobe redundancy and inefficiency, where users frequently repeat outfits or purchase unnecessary clothing. The purpose of this module is to develop an AI-powered system that enables users to make better use of their existing wardrobe while providing personalized, practical, and context-aware outfit recommendations.

The fashion and apparel industry has increasingly turned to AI-powered recommendation systems to help users make better clothing choices. Systems like Vue.ai and Dataiku's AI Outfit Recommendation System demonstrate the capability of AI to provide personalized clothing suggestions by analyzing user profiles and wardrobe data. However, most existing systems focus on recommending new purchases rather than optimizing the utilization of clothes that users already own.

The Smart Digital Wardrobe system identifies clothing type and colour from user-uploaded images, with the fabric type provided manually by the user. It then integrates user profiles, such as occupation and real-time weather data, to suggest daily outfits that are both stylish and suitable for environmental conditions.

4.2 Background and Literature Survey

Previous research and commercial solutions have explored personalized outfit suggestions, leveraging machine learning and image processing techniques to analyze user preferences, clothing attributes, and contextual information such as weather or occasion. Systems like Vue.ai and Dataiku's AI Outfit Recommendation System demonstrate the capability of AI to provide personalized clothing suggestions by analyzing user profiles and wardrobe data.

To implement a system like the Smart Digital Wardrobe, it is essential to master image recognition, clothing attribute extraction including type, colour, and fabric, and recommendation algorithms such as collaborative filtering and content-based filtering. Understanding context-aware recommendation techniques including weather integration, occupation-based personalization, and group and couple coordination is also critical. The state of the art includes AI-powered solutions capable of generating outfit suggestions based on user preferences and past behavior, but most existing systems lack group and travel wardrobe planning functionalities.

4.3 Research Gap

Despite the growing adoption of AI in fashion recommendation systems, current solutions largely focus on product-centric suggestions or trend-based recommendations, often ignoring the user's existing wardrobe. Systems like Vue.ai and Dataiku's platform primarily recommend new purchases rather than optimizing the utilization of clothes that users already own. Similarly, existing AI systems consider basic user preferences but fail to fully integrate contextual factors such as

weather conditions, user occupation, or planned trips, limiting the practical usability of recommendations.

Most commercial systems lack group or couple coordination features, which are crucial for events and social gatherings. They also do not provide mix-and-match recommendations to avoid repeated outfit usage, which is a significant issue for users seeking wardrobe efficiency and sustainable fashion practices. Current research often emphasizes algorithmic accuracy but does not comprehensively address the holistic integration of user wardrobe, context, and personalization.

4.4 Research Problem

Managing personal wardrobes efficiently is a common challenge for many users, leading to repetitive outfit choices and unnecessary clothing purchases. People often struggle to plan daily outfits based on their existing clothing, especially when considering weather conditions, occupation, or planned trips. Current fashion recommendation systems focus primarily on recommending new clothing items or following general fashion trends. Another limitation is the lack of mix-and-match suggestions and group or couple coordination, which are important for social events, trips, or family occasions.

4.5 Objectives

Main Objective: To develop an AI-powered Smart Digital Wardrobe that provides personalized, context-aware outfit recommendations using the user's existing wardrobe, improving wardrobe utilization, reducing outfit repetition, and enhancing user confidence in daily dressing.

Specific Objectives:

- Detect clothing type and colour from user-uploaded images.
- Incorporate user-provided fabric information and profile details into outfit recommendations.
- Provide weather-appropriate outfit suggestions for daily wear.
- Generate mix-and-match outfit combinations to prevent repetition and optimize wardrobe usage.
- Recommend coordinated outfits for groups or couples.
- Plan travel wardrobes, suggesting the number of outfits and daily combinations based on location, weather, and trip duration.

4.6 Methodology

4.6.1 System Architecture

The Smart Digital Wardrobe architecture consists of five integrated modules. The User Input Module allows users to upload images of clothing items and provide additional information such as fabric type, occupation, and trip details. The Data Processing Module performs image preprocessing including resizing, normalization, and cropping, and stores labeled clothing attributes in a structured database. The Clothing Recognition Module utilizes CNNs to detect clothing type and color, ensuring accurate recognition for mix-and-match recommendations. The

Recommendation Engine generates personalized daily outfit suggestions, integrates weather data, and supports group coordination and travel planning. The User Interface Module provides a responsive web interface for seamless interaction.

4.6.2 Data Collection and Preprocessing

Data collection involves gathering clothing images from online sources, fashion websites, and user uploads, alongside user inputs such as fabric type and occupation. Images are labeled with clothing attributes including type, colour, and fabric. Data preprocessing includes removing duplicates and low-quality images, resizing and normalizing images, encoding categorical attributes, and integrating multiple datasets into a structured training set.

4.6.3 Functional Requirements

- Upload clothing images and input fabric type and profile details.
- Automatically detect clothing type and color.
- Provide daily outfit suggestions with mix-and-match combinations.
- Integrate weather data for suitable outfit recommendations.
- Suggest coordinated outfits for groups or couples.
- Plan travel wardrobes using existing clothes.

4.6.4 Tools and Technologies

Programming Languages: Python, JavaScript. Machine Learning: TensorFlow, Keras, OpenCV. Web Development: Node.js, Express.js, Flask, FastAPI, React.js. UI Components: Material-UI. Database: MongoDB. Design Tools: Figma, Draw.io. Testing: Jest, PyTest, Postman. Deployment: Docker, Kubernetes. Project Management: Microsoft Planner.

4.6.5 Testing Strategy

The testing process includes unit testing and system testing among team members and friends to verify clothing recognition, mix-and-match suggestions, weather integration, and travel wardrobe planning. In the final phase, the system is tested with a selected group of students, office workers, and fashion enthusiasts in a local environment. This includes acceptance testing, with both Alpha and Beta testing, to validate usability, accuracy, and practical functionality. Most testing is manual, with users providing feedback on outfit suggestions. Internal functions and assert statements verify the correctness of clothing detection, colour recognition, and recommendation logic.

4.7 Commercialization

The target audiences are students, office workers, and fashion-conscious individuals seeking to optimize wardrobe usage and plan personalized outfits. The system can be commercialized as a web and mobile application through a subscription-based model, offering premium features such as travel wardrobe planning and group outfit coordination. Partnerships with fashion retailers, e-commerce platforms, and lifestyle apps provide additional B2C and B2B revenue

opportunities. The estimated budget includes hosting, domain, publication, and deployment costs totaling LKR 87,500.

Chapter 5: Component 3 – 3D Virtual Mannequin Generation & Body/Skin Tone Analysis

Student: Shangeetha J. (IT21188886)

5.1 Introduction

The growth of e-commerce has transformed the way consumers engage with fashion, but current virtual try-on technology is not designed to offer an inclusive, personalized, and emotionally responsive dressing experience. Current solutions are mannequin-based and rooted in static installations, do not provide realistic skin tone and body representation, and are not sensitive to differently abled users' accessibility needs. These limitations reduce realism, inclusivity, and consumer confidence and contribute to high return rates in e-commerce shopping.

Style Mate's 3D Mannequin module addresses these challenges by integrating computer vision, machine learning, and affective computing. The system offers customized 3D mannequins generated from simple 2D images, accurate skin tone analysis, emotion-aware outfit recommendations, and a chatbot to increase accessibility. The aim is to democratize fashion technology to enable users of all body shapes, cultures, and abilities to make decisions on style in an informed and confident manner.

5.2 Background and Literature Survey

The fashion industry has rapidly shifted to digital media, and Virtual Try-On (VTO) systems have emerged as promising alternatives for enabling customers to preview clothing items without physically trying them on. However, existing virtual dressing technologies typically do not provide truly personalized, realistic, and inclusive experiences. Most VTO systems use static mannequins or generic avatars that do not capture the diversity of human body shapes, poses, and cultural backgrounds.

CNNs and GANs have been utilized for clothing transfer and multi-pose outfit visualization with improved realism. Body shape classification models and 3D reconstruction techniques have been developed to create mannequins that better reflect individual user measurements. While these techniques represent important steps, they remain computationally expensive and beyond the reach of everyday users, as many still require specialized hardware such as 3D scanners.

Research in affective computing shows that incorporating user emotions, mood, and facial expressions into recommender systems can radically enhance engagement and relevance. Yet, affective fashion recommendation remains underexplored. Similarly, accessibility aspects of VTO technology are typically neglected, excluding individuals with physical or cognitive impairments.

5.3 Research Gap

Existing VTO pipelines do not have a shared, lightweight, and accessible solution that generates personalized mannequins from simple 2D images including seated and partial views, preserves skin-tone variety for realistic visualization, and

integrates affect-aware recommendations with live try-on preview. The key gaps identified include hardware dependency on expensive 3D scanners, rigid pose requirements that exclude partial and seated users, limited skin tone representation, and the absence of emotional context in recommendations.

Feature	Existing Systems	Proposed System	Gap Addressed
Hardware Requirement	High (3D Scanners, Depth Cameras)	Low (Standard RGB Camera)	Accessibility & Cost
Pose Requirement	Rigid, Standardized (T-pose)	Flexible (Seated/Standing)	Inclusivity for All
Skin Tone Consideration	Limited / Generic	Core Feature with ML Analysis	Diversity & Realism
Primary Focus	Geometric Accuracy	Inclusivity & User Experience	Market Need
Deployment	Complex, Offline Software	Web-Based, API-Driven	Scalability

5.4 Research Problem

The central research problem is the systemic exclusion caused by current VTO systems. Most systems require full-body T-poses, excluding users with mobility disabilities or situational constraints. The use of generic, often Western-centered mannequins that ignore skin tone diversity results in unrealistic visualizations for a global majority, reducing trust and contributing to fashion e-commerce return rates of 20-40%. The cost and operational complexity of 3D scanning hardware excludes SMEs and general consumers, limiting widespread adoption.

5.5 Objectives

Main Objective: To develop an end-to-end AI-driven system that generates personalized 3D virtual mannequins from standard 2D photographs, including non-standard poses, using computer vision for body shape classification and skin tone analysis, creating an accurate, inclusive, and web-deployable virtual try-on foundation.

Specific Objectives:

- Develop a secure, user-friendly web interface for image acquisition via webcam or upload, with an effective pre-processing pipeline for background elimination, normalization, and noise reduction.
- Implement a human pose estimation system using MediaPipe Holistic to extract body landmarks from pre-processed images, including seated or partial views.
- Develop anthropometric ratio computation algorithms and train a body shape classifier to categorize users into body typologies.

- Design a lighting-invariant skin tone detection pipeline using color space transformations and ML classification to classify skin tone on the Fitzpatrick or Monk scale.
- Implement a 3D mannequin generation pipeline using Blender's Python API to produce personalized, web-optimized 3D models in glTF/GLB format.
- Package the system as a containerized microservice with a RESTful API for integration with the StyleMate platform.

5.6 Methodology

5.6.1 System Architecture

The StyleMate 3D Mannequin module architecture consists of five components: the StyleMate Frontend (React/Vite) for user interaction; the 3D Mannequin Generator API (Python/Flask) as the core processing endpoint; the Image Pre-processing Module for background removal using U²-Net, normalization, and resizing; the AI Model Server (TensorFlow Serving) for pose estimation, body shape classification, and skin tone analysis; and the Data Fusion & Parameterization Module with the 3D Modeling Tool (Blender) for final avatar generation.

5.6.2 Implementation Phases

- Phase 1 – Foundation and Pre-processing: Setup development environment, install image capture/upload module, and pre-processing pipeline with background removal and normalization.
- Phase 2 – Core CV and ML Models: Implement pose estimation with MediaPipe, compute body ratios, train body shape classifier on SSP-3D dataset, and implement skin segmentation and tone classification on FairFace dataset.
- Phase 3 – 3D Integration: Implement Python scripts to control Blender, create functions to build 3D model with input parameters, and implement skin tone texture mapping.
- Phase 4 – API Development and Pipeline Integration: Integrate all modules into a single pipeline and distribute as Flask API with image input and 3D model output.
- Phase 5 – Deployment, Refinement, and Testing: Conduct unit, integration, and user acceptance testing; improve models based on feedback; containerize with Docker for cloud deployment.

5.6.3 Data Collection

Public datasets including SSP-3D for body shape and FairFace for skin tone classification are used for initial training. Synthetic data augmentation through rotation, scaling, and brightness adjustment improves model generalization. A small-scale, IRB-approved primary data collection from diverse volunteers in various poses and lighting conditions is conducted to improve model performance on real-world inputs and ensure inclusivity.

5.6.4 Evaluation Strategy

Technical accuracy is measured through body shape classification accuracy, recall, and F1-score on human-annotated test sets. Skin tone classification accuracy and equity across diverse skin tone classes are assessed. Performance is benchmarked by inference time from image input to 3D model output, with a target of under 60 seconds for a proof-of-concept. User satisfaction is measured through Mean Opinion Score (MOS) for avatar realism and System Usability Scale (SUS) for ease of use.

5.7 Commercialization

The projected 3D Virtual Mannequin Generator addresses the largest pain point in global fashion e-commerce: product returns of up to 30% for apparel. Three target market segments are identified: Large Enterprise E-Commerce Platforms seeking to reduce return rates through better fit and style preview; SMEs and D2C Brands seeking competitive differentiation; and B2B SaaS Providers looking to add high-value features to their product lines. The monetization strategy includes a Pay-Per-Use API model, tiered subscription plans, and white-label licensing for technology providers.

Chapter 6: Component 4 – Event Theme Color Analysis & Outfit Recommendation

Student: Thivvijavearl T.S (IT22269270)

6.1 Introduction

Choosing appropriate clothes for different occasions is an essential element in personal presentation and visual aesthetics. In cases such as weddings, birthday parties, business events, and themed occasions, there is a necessity to choose clothes that complement not only the particular event but also the overall environment, including the décor and color schemes. However, choosing outfits to match event themes is rather difficult since one needs to have knowledge about colors, fashions, and styles among other factors.

With the development of artificial intelligence and computer vision, there is a need to design technology-enabled intelligent models that can aid in making appropriate fashion decisions. Current fashion recommendation systems mostly consider the preferences of users, their physical appearance, seasons, and shopping history. Even though such approaches are useful, there is a drawback in that they disregard environmental contexts, especially color themes of the event décor.

This module introduces a novel AI-based Event Theme Color Analysis and Outfit Recommendation System. Through this system, a user can upload an image of the event theme decoration and the system uses image processing techniques to identify the dominant color scheme. The user then specifies the nature of the event, enabling the system to narrow down and refine the results for more contextually appropriate recommendations.

6.2 Background and Literature Survey

Artificial Intelligence integration in the fashion industry has received much interest, specifically in building intelligent fashion recommendation systems. Fashion recommendation systems help customers pick fashion products through parameters including user preference, physique, season, and history, typically adopting machine learning algorithms, collaborative filtering, and content-based filtering. Despite the extensive use of these systems, few solutions consider environmental or contextual factors.

Computer vision-based recommendation systems employ models such as CNNs to analyze images for clothing classification, style recognition, and outfit compatibility prediction. Color analysis and theory is another promising area, with principles of complementary, analogous, and triadic colors discussed in fashion design. Recently, K-Means clustering has been applied to recognize dominant colors in photographs, enabling automatic extraction of color schemes for tasks in image retrieval, branding, and interior design. Context-aware recommendation systems use additional information such as user location, weather, and activity to improve recommendation quality, though few consider visual features of the surrounding environment.

6.3 Research Gap

Despite progress in computer vision, color analysis, and recommendation systems, a significant gap remains. Almost no prior researchers have attempted to develop an algorithm that analyzes event images, extracts color palettes, and uses this information to recommend appropriate clothes. This aspect is crucial when making outfit recommendations for events.

Specifically, existing systems focus on technical aspects of image retrieval while neglecting contextual factors. Most do not leverage machine learning for intelligent image search combined with fashion recommendation. Color theory and harmony, while well-established, are not utilized by recommendation algorithms in an efficient, automated manner. Current recommendation systems are not aware of occasion-specific visual contexts. Most existing solutions address only one problem at a time rather than integrating multiple techniques.

6.4 Research Problem

The primary issue addressed by this study is the absence of an intelligent, context-sensitive fashion recommendation system capable of analyzing images from event decoration and recommending appropriate outfit colors and styles according to the thematic and visual elements and the kind of event in question. Users attending events like weddings, parties, office events, and themed gatherings often encounter difficulties choosing proper outfits which match their environment. Currently, no technological solution automates this process through analyzing event decoration pictures, determining dominant colors, and offering corresponding fashion suggestions.

6.5 Objectives

Main Objective: To create and develop an AI-powered solution capable of analyzing images of event decorations, detecting the prevailing color scheme through computer vision algorithms, and offering appropriate suggestions regarding attire colors and styles compatible with the event occasion.

Specific Objectives:

- To gather a dataset of event decoration images covering various occasions including wedding, party, corporate, and casual events.
- To build a computer vision solution for extracting dominant colors from event decoration images through image processing and K-Means clustering.
- To utilize color theory principles including complementary, analogous, and triadic color schemes for mapping extracted colors to clothing colors.
- To create an occasion-based filtering algorithm to modify outfit recommendations based on the selected event type.
- To establish recommendation logic for generating outfit suggestions for both male and female individuals using extracted color themes and fashion guidelines.

- To build a full-stack application allowing users to upload event images, choose their occasion, and get outfit recommendations instantly.
- To evaluate the system's efficiency in recommending visually suitable outfits for various occasions.

6.6 Methodology

6.6.1 Research Design

This research adopts an applied research design with emphasis on system development and implementation. The research is data-driven, using event decoration images as primary input sources. A hybrid methodology combines computer vision techniques with rule-based fashion logic: computer vision analyzes images and extracts dominant colors, while rule-based reasoning applies color harmony principles and occasion-based filtering. The study employs a prototype-based development model as a full-stack application.

6.6.2 System Architecture

The system is structured into four main layers. The Input Layer receives event decoration images and occasion type from the user. The Processing Layer performs image preprocessing, applies K-Means Clustering for dominant color extraction, and generates a structured color palette. The Recommendation Layer applies color theory principles and occasion-based filtering to generate outfit suggestions. The Application Layer provides the React.js user interface for image upload, occasion selection, and results display, with Flask backend handling all processing.

6.6.3 Data Collection and Preprocessing

The dataset was obtained from Kaggle and consists of event decoration images representing different occasions such as weddings, birthday parties, corporate events, festive celebrations, and casual gatherings. After downloading, images were organized and categorized by event type. Irrelevant, low-quality, and duplicate images were removed. All images were resized to uniform dimensions, converted to RGB format, and normalized. Color values were standardized and in some cases converted to HEX representations. Occasion labels were verified for consistency.

6.6.4 Exploratory Data Analysis

EDA revealed important patterns in the dataset. Wedding-related images typically contain lighter and more elegant color palettes including white, gold, pastel shades, and cream tones. Party-related images often include vibrant and saturated colors such as red, blue, purple, and neon variations. Corporate event images tend to use neutral and formal colors such as grey, black, and navy blue. The distribution showed that weddings and parties contained more images than other categories. Colors such as white, gold, and red appeared most frequently across the dataset.

6.6.5 Feature Engineering

Feature engineering combines three types of features. Structured Color Features consist of dominant colors extracted from each image using K-Means clustering, represented as RGB and HEX values forming a color palette feature vector. Color

Harmony Features are derived from color theory principles, including complementary colors (opposite on the color wheel), analogous colors (adjacent, creating smooth harmony), and triadic colors (evenly spaced, creating balanced contrast). Occasion-Based Features consist of categorical labels for each event type, such as wedding, party, corporate, and casual, which influence the final recommendation output. The final hybrid feature representation combines all three types for both visual and contextual intelligence.

6.6.6 Model Development

The system employs a hybrid rule-based and color-theory-driven approach. The Color Extraction Model uses K-Means clustering on image pixel data to identify dominant colors, representing each image as a color palette. The Color Matching and Harmony Model maps extracted colors to suitable clothing colors using complementary matching for strong contrast, analogous matching for harmonious appearance, and neutral balancing for complex backgrounds. The Occasion-Based Decision Model adjusts outfit style and formality based on the selected event: weddings generate elegant and traditional suggestions, corporate events yield professional and minimal palettes, parties allow bold and vibrant choices, and casual events produce relaxed and flexible recommendations. A Rule-Based Recommendation Engine combines both models to generate final outfit suggestions for men and women separately.

6.6.7 System Implementation

The frontend is developed using React.js with an image upload interface, occasion selection dropdown, real-time display of results, color palette visualization, and display of recommended outfit colors for men and women. The backend is developed using Python Flask and handles image reception, preprocessing, K-Means clustering for color extraction, color mapping using color theory rules, occasion-based filtering, and recommendation generation. The frontend and backend communicate via RESTful APIs: the user uploads an image and selects an occasion, the frontend sends an HTTP request to the Flask backend, the backend processes the image and returns a JSON response containing the dominant color palette, recommended outfit colors, and gender-specific suggestions, and the frontend displays the results.

6.6.8 Testing and Evaluation

Functional testing verified that image upload, occasion selection, dominant color extraction, API communication, and result display all operate as intended. Color extraction accuracy testing on multiple event decoration images showed that K-Means clustering successfully identified prominent colors in most images, with minor variations in cases of highly complex backgrounds or mixed lighting. Recommendation quality was evaluated based on color harmony accuracy, occasion relevance, and visual suitability, with the system successfully generating contextually appropriate outfit suggestions. User experience evaluation confirmed that the React-based interface provided simple and intuitive interaction with real-time response.

6.7 Conclusion

The Event Theme Color Analysis and Outfit Recommendation System successfully integrates computer vision and fashion intelligence to provide context-aware outfit recommendations based on event decoration images. By combining K-Means clustering for dominant color extraction, color theory for harmony matching, and occasion-based filtering, the system generates meaningful and visually appropriate recommendations that go beyond the capabilities of traditional user-centric fashion systems. Future work includes integration of deep learning for advanced image understanding, addition of personalized user preferences, a larger and more diverse dataset, a mobile application, and integration with virtual try-on technology.

Chapter 7: Individual Contributions

7.1 Contribution Summary

Module	Student	IT Number	Key Contribution
Accessory Recommendation & Customization	Vishnuyan T.	IT22208194	Computer vision-based attribute extraction; rule-based accessory recommendation engine; user-driven customization module
Smart Digital Wardrobe	Abirajh A.	IT22066848	Clothing image recognition; wardrobe management system; weather-aware and mix-and-match outfit generation
3D Virtual Mannequin & Body/Skin Tone Analysis	Shangeetha J.	IT21188886	Pose estimation pipeline; body shape classification; skin tone detection; 3D mannequin generation using Blender API
Event Theme Color Analysis & Outfit Recommendation	Thivvijavearl T.S	IT22269270	Event image analysis; K-Means dominant color extraction; color theory-based outfit mapping; occasion-based filtering

7.2 Component 1 – Accessory Recommendation & Customization (Vishnuyan T.)

Vishnuyan T. was responsible for the full design and development of the Accessory Recommendation and Customization System. His key contributions included designing and implementing the computer vision pipeline for extracting body shape, face shape, dress color, neckline style, and skin tone attributes from user-uploaded full-body images using TensorFlow and OpenCV. He developed the rule-based recommendation engine that applies fashion styling rules to suggest contextually appropriate accessories. He also implemented the user-driven customization module using PyTorch-based generative models and the MongoDB accessory catalog. Additionally, he conducted risk analysis, designed the testing strategy, developed the commercialization plan, and prepared the individual research proposal.

7.3 Component 2 – Smart Digital Wardrobe (Abirajh A.)

Abirajh A. led the development of the Smart Digital Wardrobe module. His contributions include designing and implementing the clothing image recognition system using CNNs with TensorFlow and Keras, and building the wardrobe management database with MongoDB. He developed the mix-and-match outfit generation algorithm, integrated the OpenWeather API for real-time weather-based recommendations, and implemented group and couple outfit coordination features. He also built the travel wardrobe planning component and conducted feasibility studies, requirement gathering, and budget planning for the module.

7.4 Component 3 – 3D Virtual Mannequin Generation (Shangeetha J.)

Shangeetha J. designed and implemented the 3D Virtual Mannequin Generation module. Her contributions include implementing the image pre-processing pipeline with background removal using U²-Net, configuring the MediaPipe Holistic pose estimation model for landmark extraction, and developing the anthropometric ratio computation algorithms. She trained and validated the body shape classifier using the SSP-3D dataset, implemented the skin tone segmentation and classification pipeline using CIELAB color space on the FairFace dataset, and developed the Blender Python API integration for 3D mesh generation and skin texture application. She also conducted the feasibility study, system evaluation design, and commercialization strategy for the module.

7.5 Component 4 – Event Theme Color Analysis (Thivvijavearl T.S)

Thivvijavearl T.S. developed the Event Theme Color Analysis and Outfit Recommendation System. His contributions include collecting and preprocessing the Kaggle event decoration image dataset, implementing the K-Means clustering algorithm for dominant color extraction, and developing the color theory-based mapping engine using complementary, analogous, and triadic color principles. He designed the occasion-based filtering mechanism, implemented the rule-based recommendation engine generating gender-specific outfit suggestions, built the Flask API backend and React.js frontend for the module, and conducted all functional testing and evaluation of recommendation quality.

Chapter 8: Results and Discussion

8.1 Component-wise Results

8.1.1 Accessory Recommendation System

The Accessory Recommendation and Customization System demonstrated successful extraction of wearer attributes from user-uploaded images under standard lighting and image quality conditions. The rule-based recommendation engine consistently generated contextually appropriate accessory suggestions aligned with extracted attributes. The customization module successfully generated and retrieved user-specified accessory designs. Testing confirmed that the system meets its functional requirements and achieved the target accuracy for attribute extraction in controlled conditions. Areas for improvement were identified in cases of unusual lighting or image occlusion.

8.1.2 Smart Digital Wardrobe

The Smart Digital Wardrobe system successfully recognized clothing types and colors from user-uploaded images, with the CNN model performing well across diverse clothing categories. The mix-and-match recommendation engine generated diverse and non-repetitive outfit combinations, and the weather integration provided contextually relevant suggestions. Group and couple coordination features were validated through user testing, receiving positive feedback for usability. Travel wardrobe planning functionality successfully generated appropriate outfit plans based on trip duration and weather data.

8.1.3 3D Virtual Mannequin Generation

The 3D mannequin generation module successfully produced personalized avatars from standard 2D images, including seated and partially occluded inputs. Body shape classification achieved good accuracy on the test set, and skin tone classification demonstrated equitable performance across diverse skin tone classes. The generated 3D models were successfully exported in glTF/GLB format and rendered in the frontend interface. User satisfaction scores in preliminary testing indicated positive perception of avatar realism and accuracy.

8.1.4 Event Theme Color Analysis

The Event Theme Color Analysis system successfully extracted dominant color palettes from event decoration images across all tested occasion categories. K-Means clustering accurately identified the most significant colors in well-lit, clear decoration images. Color theory-based outfit mapping produced visually harmonious recommendations, and occasion-based filtering correctly adjusted suggestion formality for different event types. Gender-specific outfit recommendations were successfully generated and displayed through the React.js interface. Minor limitations were observed in images with highly complex or mixed-lighting backgrounds.

8.2 Research Findings and Discussion

The StyleMate system demonstrates that the integration of multiple specialized AI modules into a unified fashion assistant produces meaningful improvements over

single-domain fashion recommendation systems. Several key findings emerge from the combined results.

The combination of visual attribute extraction and rule-based reasoning provides a practical and interpretable alternative to fully data-driven models, particularly in domains where labeled training data is limited, such as accessory recommendation. The hybrid approach offers explainability, which is important for user trust in fashion recommendations.

Contextual factors, including event themes, weather conditions, user occupation, and occasion type, significantly improve the relevance and usability of fashion recommendations. Systems that incorporate environmental context outperform user-preference-only systems in real-world styling scenarios.

The inclusive design principles embedded in the 3D mannequin module, particularly the support for non-standard poses and diverse skin tones, demonstrate the feasibility of accessible virtual try-on technology without requiring specialized hardware. This validates the research direction of lightweight, software-only avatar generation.

User feedback indicated that the integrated platform provides a more complete and satisfying styling experience than individual components in isolation, confirming the value of the modular yet interconnected architecture of StyleMate.

Chapter 9: Conclusion and Future Work

9.1 Conclusion

This research presented the design and implementation of StyleMate, an integrated AI-powered virtual fashion assistant comprising four specialized modules: Accessory Recommendation and Customization, Smart Digital Wardrobe, 3D Virtual Mannequin Generation, and Event Theme Color Analysis and Outfit Recommendation. Each module addresses a distinct gap in existing virtual fashion technology, and together they form a comprehensive system capable of delivering personalized, context-aware, and inclusive styling guidance.

The project successfully demonstrated that computer vision, machine learning, color theory, and rule-based reasoning can be effectively combined to solve real-world fashion decision-making challenges. The unified full-stack architecture enables seamless data sharing between modules and delivers a consistent user experience through a single web application interface.

Across all four modules, the research confirms that incorporating contextual factors, including environmental color themes, weather conditions, user profiles, body characteristics, and occasion type, significantly enhances the relevance and quality of fashion recommendations compared to traditional product-centric approaches. The system's inclusive design principles further demonstrate the potential for AI-driven fashion technology to serve diverse user populations.

9.2 Future Work

- Integration of deep learning-based models, including transformer-based vision models, to improve accuracy in complex image conditions across all modules.
- Addition of personalized user preference learning, enabling the system to adapt recommendations based on feedback history and interaction patterns.
- Development of a mobile application version to allow real-time event image capture and instant outfit recommendations.
- Integration of virtual try-on technology with the 3D mannequin module, allowing users to visualize complete outfit recommendations on their personalized avatar.
- Expansion of the dataset to include more diverse global event styles, body types, and fashion contexts for improved generalization.
- Incorporation of emotion-aware and mood-based recommendation features using affective computing to further personalize the styling experience.
- Implementation of a social sharing feature allowing users to share outfit recommendations and wardrobe combinations with friends or fashion communities.

Budget

Category	Description	Estimated Cost (LKR)
Hosting & Cloud	Basic cloud server / GPU rental for 6 months	30,000.00
Domain & Web Hosting	Domain (stylemate.lk) + basic web hosting	10,000.00
Publication	Conference/journal fees + printing/binding	12,500.00
Deployment	Cloud deployment of prototype (3–4 months)	15,000.00
Marketing (Optional)	Posters, brochures, small-scale promotions	20,000.00
Total Estimated		87,500.00

Note: Costs may vary depending on economic conditions. GPU rental costs may increase based on model complexity and training duration.

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